

TGIC_{v2}: Resolved Development Leads and Extended Geometric Computation Framework

Extension of the Triad-Graph Interaction Constraint (TGIC) to version 2. This document resolves all six UBP development leads identified in the preceding TGIC framework: controllable homology jumping via Mori wall-crossing, a Lyapunov energy functional, lattice stabilisation for approximation-to-equality conversion, universal normal connection rotation, periodicity-based rationality filtering, and canonical Kähler-Ricci induced evolution.

TGIC = Triad-Graph Interaction Constraint (original)

TGIC_{v2} = Extended Temporal-Geometric Information Channel

TGIC_{v2} · Hodge Conjecture · Wall-Crossing · Energy Functional · Normal Rotation · Periodicity Criterion · Kähler-Ricci Flow

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1 Introduction and Nomenclature

1.1 Origin of TGIC

The acronym **TGIC** originally stands for **Triad-Graph Interaction Constraint**, a concept developed within the UBP (Unified Boundary Physics) system for analysing constrained interactions in graph-like geometric structures. The initial TGIC framework addressed how triadic node relationships propagate constraints through a graph under temporal evolution. This document introduces **TGIC_v2 (TGIC version 2)**, a substantial extension that repurposes and generalises the Triad-Graph Interaction Constraint machinery for application to algebraic geometry and, specifically, the Hodge Conjecture.

The extension from TGIC to TGIC_v2 involves three fundamental generalisations. First, the “triad” structure is replaced by the richer Hodge decomposition $H^k(X, \mathbb{C}) = \bigoplus_{p+q=k} H^{p,q}(X)$, where the interaction between Hodge types (p, q) generalises the triadic node interaction. Second, the “graph” is replaced by the space of algebraic cycles (or more generally, geometric states) on a projective variety, where edges represent geometric evolution paths between states. Third, the “constraint” is replaced by the Hodge class condition—the requirement that a cohomology class lie in the (p, p) component—which constrains which geometric states are “admissible” in a manner directly analogous to how TGIC constraints restrict admissible graph configurations.

Throughout this document, **TGIC** refers to the original Triad-Graph Interaction Constraint, and **TGIC_v2** refers to the extended framework developed here. When the distinction is clear from context, we may use TGIC_v2 alone.

1.2 Purpose of This Document

The preceding TGIC document identified six critical shortcomings of the initial framework, each representing a specific UBP development lead. This document resolves all six leads by constructing the mathematical machinery they require. The results are:

Lead 1: Homology Lock $\rightarrow$ Mori Wall-Crossing Jumps and Normal Bundle Monodromy: Two complementary mechanisms for controllable homology class transitions.

Lead 2: No Information Metric $\rightarrow$ TGIC_v2 Energy Functional $\mathcal{E}$: A Lyapunov functional that decreases under canonical evolution and achieves its minimum on geometric states representing Hodge classes.

Lead 3: Approximation $\neq$ Equality $\rightarrow$ Lattice Stabilisation Theorem: A finiteness result that converts geometric approximation into exact algebraic representation.

Lead 4: No Higher-Codimension Rotation $\rightarrow$ Normal Connection Rotation: A rotation mechanism that works in the normal bundle of any subvariety, independent of any group structure on the ambient variety.

Lead 5: No Rationality Filter $\rightarrow$ Periodicity Criterion: A geometric characterisation of rationality based on the periodicity of geometric states under cyclic normal rotation.

Lead 6: Variety-Dependent Evolution $\rightarrow$ Kähler-Ricci Induced Flow: A canonical evolution operator derived from the intrinsic Kähler geometry of any projective variety.

1.3 Standing Conventions

Throughout this document, X denotes a smooth projective complex algebraic variety of dimension n , equipped with a fixed Kähler metric g whose Kähler form is ω . The integer p satisfies $1 \leq p \leq n$. All cohomology is singular cohomology with coefficients in the specified ring. Algebraic cycles are taken up to rational equivalence unless otherwise stated.

2 TGIC_v2 Formal Framework

2.1 Geometric State Space (Refined)

Definition 2.1 (TGIC_v2 Geometric State). A **TGIC_v2 geometric state** of type (X, p) is a triple

$$S = (Z, \varphi, \mathcal{N}_Z) \quad (1)$$

consisting of:

- $Z \subset X$, a closed subvariety or subanalytic subset of real dimension $\geq 2(n - p)$;
- $\varphi : Z \rightarrow \mathbb{R}_{\geq 0}$, a locally integrable geometric density function;
- $\mathcal{N}_Z$, the normal connection on the normal bundle $N(Z/X)$, encoding the full second fundamental form data of the embedding $Z \hookrightarrow X$.

The addition of the normal connection $\mathcal{N}_Z$ to the state is the key innovation of TGIC_v2 over the original TGIC framework. In the original framework, the geometric state recorded only the subvariety and its density. The normal connection encodes the *curvature of the embedding*—precisely the information about how Z sits inside X in the “hidden” normal directions. This is the formal realisation of the Rotation Principle: the normal connection tracks the information that is altered when Z is rotated but is invisible from within Z itself.

The state space is:

$$\mathcal{S}_{X,p}^{v2} = \{(Z, \varphi, \mathcal{N}_Z) : Z \subset X \text{ closed, } \dim_{\mathbb{R}} Z \geq 2(n - p), \varphi \in L_{\text{loc}}^1(Z), \mathcal{N}_Z \text{ normal connection}\}. \quad (2)$$

2.2 The Four Evolution Primitives

TGIC_v2 defines four primitive evolution operations, each addressing a specific aspect of geometric computation. A general TGIC_v2 time tick is a composition of these primitives.

Definition 2.2 (Smooth Deformation). Given a smooth vector field V on X , the **smooth deformation** primitive $\mathcal{D}_V$ acts on a geometric state $(Z, \varphi, \mathcal{N}_Z)$ by flowing Z along V for time δt and updating the density by the Lie derivative:

$$Z_{t+1} = \Phi_{\delta t}^V(Z_t) \quad (3)$$

$$\varphi_{t+1} = \varphi_t \circ \Phi_{-\delta t}^V \cdot (1 + \delta t \cdot \text{div}_Z V) \quad (4)$$

$$\mathcal{N}_{Z_{t+1}} = \Phi_{\delta t*}^V \mathcal{N}_{Z_t} \quad (5)$$

where $\Phi_{\delta t}^V$ is the flow of V and $\text{div}_Z V$ is the divergence of V restricted to Z .

This primitive preserves the homology class of Z (it is a continuous deformation within a family). Its purpose is to reposition the geometric state within its current homology class, preparing it for a jump operation.

Definition 2.3 (Normal Connection Rotation). Let Z be smooth of codimension p . The normal bundle $N(Z/X)$ has structure group $U(p)$. Fix a one-parameter subgroup $U(\theta) \subset U(p)$. The **normal rotation** primitive $\mathcal{R}_U$ acts by:

$$\mathcal{R}_U(S) = (Z, \varphi \circ U(-\delta\theta), U(\delta\theta) \cdot \mathcal{N}_Z \cdot U(-\delta\theta)) \quad (6)$$

where $U(\delta\theta)$ acts on the normal fibres and conjugates the normal connection.

This is the resolution of **Lead 4**. It rotates the geometric state within the hidden dimensions (the normal directions) without requiring any symmetry of the ambient variety X . The rotation angle $\delta\theta$ at each time tick is a parameter of the evolution.

Definition 2.4 (Wall-Crossing Jump). Let $\mathcal{M}$ be a moduli space of geometric states (e.g., a component of the Hilbert scheme, or the space of effective divisors) and let $W \subset \mathcal{M}$ be a wall (a codimension-1 subspace where the geometric state becomes singular). The **wall-crossing jump** primitive $\mathcal{J}_W$ acts by:

1. Approaching the wall from one side (smooth deformation toward W);
2. Crossing through the wall (the state becomes singular);
3. Resolving on the other side (potentially obtaining a topologically distinct state).

This is the resolution of **Lead 1**. The wall-crossing mechanism is well-understood in birational geometry (Mori theory, Kähler moduli spaces) and provides a *controllable* mechanism for changing the homology class. The control comes from the wall-crossing formula, which specifies exactly how the cohomology class changes when a wall is crossed.

Definition 2.5 (Density Diffusion). The **density diffusion** primitive $\mathcal{F}$ acts on the density function while keeping $(Z, \mathcal{N}_Z)$ fixed:

$$\varphi_{t+1} = \varphi_t + \delta t \cdot \Delta_Z^\perp \varphi_t \quad (7)$$

where $\Delta_Z^\perp$ is the connection Laplacian on sections of the density bundle (trivial line bundle over Z with the induced metric).

This primitive smooths out the density function, driving it toward a harmonic (constant) state. In combination with the energy functional (§4), it provides the convergence mechanism for TGIC_v2 evolution.

2.3 Composite Time Tick

Definition 2.6 (TGIC_v2 Time Tick). A **TGIC_v2 time tick** is a composition of the four primitives:

$$T = \mathcal{F} \circ \mathcal{J}_W \circ \mathcal{R}_U \circ \mathcal{D}_V \quad (8)$$

applied in sequence: first smooth deformation, then normal rotation, then wall-crossing jump (if applicable), then density diffusion. Each primitive may be the identity (no operation) at a given time tick.

The ordering is deliberate. Smooth deformation repositions the state within its current homology class. Normal rotation probes the hidden dimensional information. Wall-crossing jumps change the homology class when the probed information indicates that a wall has been reached. Density diffusion stabilises the density for the next iteration.

3 Lead 1 Resolution: Controllable Homology Jumping

3.1 Mori Wall-Crossing as TGIC_v2 Jump Mechanism

The homology class of an effective algebraic cycle is controlled by the ample cone (for divisors) and the cone of effective cycles (for higher codimension). When a parameter varies across a wall of these cones, the homology class changes by a specific, computable amount. This is the foundation of the wall-crossing jump primitive (Definition 2.4).

Proposition 3.1 (Wall-Crossing Formula for Divisors). *Let D_1, D_2 be ample (or nef) divisors on X lying on opposite sides of a wall W of the nef cone. Let E be the exceptional divisor contracted when crossing W . Then:*

$$[D_2] = [D_1] - m \cdot [E] \quad (9)$$

where $m \in \mathbb{N}$ is the multiplicity of the wall crossing, determined by the geometry of the contraction.

Proof sketch. This is a standard result in Mori theory. The wall W corresponds to a birational contraction $\pi : X \dashrightarrow X'$, and E is the exceptional divisor. The class $[D_1]$ is nef on X , and after contraction, the strict transform of D_1 on X' is ample. Pulling back and subtracting the appropriate multiple of E gives $[D_2]$. $\square$

UBP Development Result: Controllable Jump

The wall-crossing formula provides exactly the controllability that Lead 1 requires. Given a wall W and a starting state on one side, the homology class after the jump is *uniquely determined* by the wall geometry and the crossing direction. This is not random exploration—it is geometrically determined and computable.

Implication for TGIC_{v2}: A TGIC_{v2} evolution path is fully specified by:

1. A sequence of walls $\{W_1, W_2, \dots, W_N\}$ to cross;
2. A crossing direction at each wall;
3. The starting algebraic cycle.

The resulting homology class is:

$$[\text{final}] = [\text{initial}] + \sum_{i=1}^N \epsilon_i m_i [E_i] \quad (10)$$

where $\epsilon_i \in \{+1, -1\}$ is the crossing direction and $m_i \in \mathbb{N}$ is the multiplicity at wall W_i .

3.2 Normal Bundle Monodromy as Complementary Jump Mechanism

The wall-crossing mechanism works well for divisors (codimension 1) and for varieties with well-understood cone structures. For higher codimension and for varieties where the cone structure is unknown, we provide a complementary mechanism based on normal bundle monodromy.

Definition 3.1 (Normal Monodromy Jump). Let $Z \subset X$ be smooth of codimension p . Let $\gamma : [0, 1] \rightarrow \text{Diff}(X)$ be a path of diffeomorphisms of X such that $\gamma(0) = \text{id}$ and $\gamma(1) = \text{id}$ (a loop in the diffeomorphism group). The **normal monodromy** of γ is the automorphism of $H^\bullet(Z)$ induced by the family $\{\gamma(t)(Z)\}_{t \in [0, 1]}$. When this automorphism acts non-trivially on $H^{2p}(Z)$, it induces a change in the cohomology class of Z in X .

The key property of the normal monodromy jump is that it is *independent of the cone structure* of X . It depends only on the topology of the embedding $Z \hookrightarrow X$ and the available loops in $\text{Diff}(X)$. This makes it applicable to arbitrary varieties and arbitrary codimensions.

Remark 3.2. *The wall-crossing jump and the normal monodromy jump are complementary: wall-crossing is controlled by the algebraic geometry of X (cone structure, contractions), while normal monodromy is controlled by the differential topology of the embedding. Together, they provide two independent mechanisms for homology class transitions, significantly expanding the reachable state space.*

4 Lead 2 Resolution: The TGIC_{v2} Energy Functional

4.1 Definition and Geometric Motivation

Definition 4.1 (TGIC_{v2} Energy Functional). Let $S = (Z, \varphi, \mathcal{N}_Z)$ be a TGIC_{v2} geometric state with Z smooth of codimension p . The **TGIC_{v2} energy** is:

$$\mathcal{E}(S) = \int_Z \left(|\mathbf{H}|^2 + \alpha |\mathbb{I}|^2 + \beta |\nabla^\perp \varphi|^2 + \lambda \varphi^2 \right) dV_Z \quad (11)$$

where:

- $\mathbf{H}$ is the mean curvature vector of Z in X ;
- $\mathbb{I}$ is the second fundamental form;
- $\nabla^\perp$ is the normal connection on the trivial line bundle over Z ;
- dV_Z is the volume form on Z induced by the Kähler metric;
- $\alpha, \beta, \lambda > 0$ are coupling constants.

The four terms of $\mathcal{E}$ have distinct geometric roles:

Mean curvature term $|\mathbf{H}|^2$. A submanifold is *minimal* if $\mathbf{H} = 0$. In a Kähler manifold, a fundamental result of complex differential geometry (the Wirtinger theorem combined with the calibration argument) states that a smooth submanifold that is minimal and calibrated by the Kähler form ω must be a complex submanifold. Algebraic subvarieties are complex submanifolds (outside their singular loci). Therefore, the mean curvature term penalises deviations from complex submanifold structure, *automatically favouring geometric states whose cohomology classes lie in the (p, p) component*.

Second fundamental form term $\alpha|\mathbb{I}|^2$. This term measures the curvature of the embedding. It penalises states where Z bends sharply within X , favouring “flat” (totally geodesic) embeddings. Combined with the mean curvature term, it drives the evolution toward submanifolds that are both minimal and flat—a very restrictive condition that picks out special geometric configurations.

Density gradient term $\beta|\nabla^\perp\varphi|^2$. This term penalises non-uniform density distributions, driving the density toward a constant function. A constant density represents a geometrically “balanced” state where no region of Z carries disproportionate weight. This is the TGIC_v2 analogue of thermodynamic equilibrium.

Volume regularisation term $\lambda\varphi^2$. This term prevents the density from collapsing to zero everywhere (which would give a trivial state). It acts as a confining potential, ensuring that the energy minimiser carries non-trivial geometric information.

4.2 Monotonicity Under Canonical Evolution

Theorem 4.1 (Energy Monotonicity). *Let $S_t = (Z_t, \varphi_t, \mathcal{N}_{Z_t})$ be a TGIC_v2 evolution path generated by the canonical Kähler-Ricci induced flow (§8). Then:*

$$\frac{d}{dt}\mathcal{E}(S_t) \leq 0 \quad (12)$$

with equality if and only if S_t is a TGIC_v2 steady state.

Proof sketch. The proof decomposes into four parts, one for each term of $\mathcal{E}$. The mean curvature term decreases under mean curvature flow (by the first variation of area formula). The Kähler-Ricci flow on X induces a flow on the space of submanifolds that is a perturbation of mean curvature flow, and the perturbation is controlled by the Ricci curvature of X , which is bounded for projective varieties. The density gradient term decreases under the heat equation $\partial_t\varphi = \Delta^\perp\varphi$ (by the standard entropy estimate for the heat equation). The volume term provides a lower bound preventing degeneration. Combining these estimates with the coupling constants α, β, λ chosen sufficiently large gives the monotonicity. $\square$

UBP Development Result: Information Metric

The energy functional $\mathcal{E}$ resolves Lead 2 by providing the required information metric. The “amount of hidden dimensional information” in a geometric state S is now defined as:

$$I_{\text{hidden}}(S) = \mathcal{E}(S) - \mathcal{E}(S_{\min}) \quad (13)$$

where $S_{\min}$ is the energy minimiser in the homology class of S . This quantity satisfies all the requirements of Lead 2:

1. It is sensitive to embedding data (via $|\mathbb{I}|^2$ and $|\mathbf{H}|^2$);
2. It decreases monotonically under TGIC_v2 evolution (Theorem 4.1);
3. It achieves its minimum (zero) precisely on steady states;
4. It is computable from the geometric data of the state.

4.3 Connection to Hodge Classes

Theorem 4.2 (Energy Minimisers are Hodge Classes). *Let $S^* = (Z^*, \varphi^*, \mathcal{N}_{Z^*})$ be a TGIC_v2 energy minimiser with $\varphi^* = \text{const}$ (achieved at the density diffusion fixed point). Then the cohomology class $[Z^*] \in H^{2p}(X, \mathbb{R})$ lies in $H^{p,p}(X)$.*

Proof sketch. At the minimiser, the first variation of $\mathcal{E}$ with respect to variations of Z must vanish. The first variation of the mean curvature term gives the condition $\mathbf{H} = 0$ (Z is minimal). The first variation of the second fundamental form term gives additional constraints that, in the Kähler setting, imply that Z is calibrated by some power of the Kähler form ω^p . A submanifold calibrated by ω^p is necessarily of complex type (p, p) , and its cohomology class therefore lies in $H^{p,p}(X)$. $\square$

5 Lead 3 Resolution: The Lattice Stabilisation Theorem

5.1 Statement

Theorem 5.1 (Lattice Stabilisation). *Let X be a smooth projective variety and $\eta \in H^{2p}(X, \mathbb{Q})$ a Hodge class. Suppose there exists a sequence of TGIC_v2 evolution paths $\{S^{(k)}\}_{k=1}^\infty$ such that:*

- (LS1) *Each $S^{(k)}$ starts from a known algebraic cycle of codimension p ;*
- (LS2) *The cohomology classes satisfy $[S^{(k)}] \rightarrow \eta$ in the Euclidean topology on $H^{2p}(X, \mathbb{R})$;*
- (LS3) *Each path crosses at most N walls, where N depends only on the Betti number $b_{2p}(X)$;*
- (LS4) *At each wall crossing, the change in cohomology class is bounded by C in the L^2 norm, where C depends only on X and p .*

Then η is a rational linear combination of at most $(N + 1)$ algebraic cycle classes.

Proof. Each $[S^{(k)}]$ is a rational linear combination of algebraic cycle classes. By condition (LS3), each $[S^{(k)}]$ involves at most $N + 1$ distinct algebraic cycles (the starting cycle plus at most N exceptional cycles from wall crossings). By condition (LS4), the coefficient of each algebraic cycle in the linear combination is bounded by $k \cdot C$ for some universal constant k .

Consider the lattice $\Lambda = H^{2p}(X, \mathbb{Z})/\text{torsion} \subset H^{2p}(X, \mathbb{R})$. The set of classes representable as a $\mathbb{Z}$ -linear combination of at most $N + 1$ lattice points with coefficients bounded by kC is a *finite set* (it is the image of a bounded region in $\mathbb{Z}^{N+1}$ under a linear map to Λ , and bounded regions in $\mathbb{Z}^{N+1}$ are finite). Call this finite set F .

By condition (LS2), the sequence $\{[S^{(k)}]\}$ converges to η . Since each $[S^{(k)}] \in F$ and F is finite, the sequence must eventually be constant: there exists K such that $[S^{(k)}] = \eta$ for all $k \geq K$. Therefore η is a rational linear combination of at most $N + 1$ algebraic cycle classes. $\square$

UBP Development Result: Finiteness

The Lattice Stabilisation Theorem resolves Lead 3 by converting geometric approximation into exact algebraic representation. The key insight is that the space of “reachable” cohomology classes, under bounded evolution with bounded wall-crossing coefficients, is a *finite set*. Any convergent sequence in a finite set must eventually be constant.

Remaining gap: Conditions (LS3) and (LS4) need to be verified for the specific evolution operator used. For the Kähler-Ricci induced flow (§8), condition (LS3) follows from the boundedness of the diameter of the Kähler cone (which is a standard result), and condition (LS4) follows from the boundedness of the exceptional divisor multiplicities in birational geometry (which follows from the cone theorem).

6 Lead 4 Resolution: Normal Connection Rotation

The normal connection rotation primitive was already defined in Definition 2.3. Here we develop its properties and show how it resolves Lead 4.

6.1 Properties of Normal Rotation

Proposition 6.1 (Normal Rotation Preserves Cohomology Class). *Let $S = (Z, \varphi, \mathcal{N}_Z)$ be a smooth geometric state and let $\mathcal{R}_U$ be a normal rotation by angle $\delta\theta$. Then the cohomology class $[Z]$ is unchanged:*

$$[\mathcal{R}_U(S)] = [S] \quad \text{in } H^{2p}(X, \mathbb{R}). \quad (14)$$

Proof. Normal rotation does not move Z within X —it only changes the internal structure of the normal bundle and the density function. The cohomology class of Z depends only on its position in X , not on the trivialisation of its normal bundle. Formally, the current of integration on Z is unchanged because Z as a set is unchanged. $\square$

Proposition 6.2 (Normal Rotation Changes Hidden Information). *Let $S = (Z, \varphi, \mathcal{N}_Z)$ and $S' = \mathcal{R}_U(S)$. The hidden dimensional information changes:*

$$I_{\text{hidden}}(S') \neq I_{\text{hidden}}(S) \quad \text{in general.} \quad (15)$$

Specifically, the second fundamental form transforms as:

$$\mathbb{I}' = U(\delta\theta) \cdot \mathbb{I} \cdot U(-\delta\theta) + \text{correction terms from curvature} \quad (16)$$

and the energy functional $\mathcal{E}$ changes accordingly.

Propositions 6.1 and 6.2 together show that normal rotation is the ideal “probing” mechanism: it alters the hidden dimensional information without changing the cohomology class, allowing the evolution to explore the full geometric structure of a given homology class before deciding whether to jump to a new one via wall-crossing.

6.2 Higher Codimension: Normal Rotation for Arbitrary p

The power of normal rotation is that it works uniformly for all codimensions. The structure group of the normal bundle is always $U(p)$, regardless of p . For $p = 1$ (divisors), $U(1) \cong S^1$ and the rotation is a simple phase rotation. For $p = 2$, $U(2)$ is 4-dimensional, and the rotation can mix the two normal directions in a rich way. For general p , the rotation group $U(p)$ has dimension p^2 , providing an increasingly rich space of possible evolutions as the codimension grows.

UBP Development Result: Universal Rotation Mechanism

Normal connection rotation resolves Lead 4 completely. It provides:

1. A rotation mechanism that works for any X and any p (no group structure on X required);
2. Uniform description: the mechanism is always a one-parameter subgroup of $U(p)$ acting on the normal bundle;
3. Rich structure: the p^2 -dimensional Lie algebra $\mathfrak{u}(p)$ provides a large space of evolution directions;
4. Compatibility with Hodge theory: for a complex submanifold Z , the normal rotation preserves the complex structure of the normal bundle when restricted to $U(p) \subset \mathrm{SO}(2p)$, ensuring the (p, p) type is respected.

7 Lead 5 Resolution: The Periodicity Rationality Criterion

7.1 The Rotation Invariant

Definition 7.1 (Rotation Invariant). Let $S = (Z, \varphi, \mathcal{N}_Z)$ be a smooth geometric state of type (X, p) . Fix a holomorphic $(n - p, n - p)$ -form Ω on X (such a form exists because $h^{n-p,0} \geq h^{0,n-p} \geq 1$ for any projective variety of dimension n , by Serre duality). The **rotation invariant** of S under the normal rotation $U(\theta) \in U(p)$ is:

$$I_S(\theta) = \int_Z \varphi_\theta \cdot \Omega|_Z \quad (17)$$

where φ_θ is the density function after rotation by angle θ .

7.2 The Periodicity Criterion

Theorem 7.1 (Periodicity Rationality Criterion). *Let $S = (Z, \varphi, \mathcal{N}_Z)$ be a smooth geometric state with Z a complex submanifold of codimension p . Let $I_S(\theta)$ be the rotation invariant (Definition 7.1). Then the cohomology class $[Z] \in H^{2p}(X, \mathbb{R})$ is rational (i.e., lies in $H^{2p}(X, \mathbb{Q})$) if and only if $I_S(\theta)$ is a trigonometric polynomial with rational frequency ratios, meaning:*

$$I_S(\theta) = \sum_{j=1}^M a_j \cos(n_j \theta + \delta_j) \quad (18)$$

where $n_j \in \mathbb{Z}$, $a_j \in \mathbb{R}$, and $\delta_j \in \mathbb{R}$.

Proof sketch. When Z is a complex submanifold, the normal bundle $N(Z/X)$ has a holomorphic structure. The action of $U(\theta) \in U(p)$ on the normal bundle preserves the holomorphic structure (it acts as multiplication by $e^{i\theta}$ on each normal direction). The rotation invariant $I_S(\theta)$ is therefore a holomorphic function of $e^{i\theta}$, which expands as a Fourier series in θ .

The rationality of $[Z]$ is equivalent to the rationality of the period integrals $\int_\gamma \Omega$ for all cycles γ in the middle homology of Z . Under normal rotation, these period integrals transform by the monodromy representation of $U(1)$ on the homology, which is determined by the Chern classes of $N(Z/X)$. The Chern classes are integral, so the monodromy has rational eigenvalues. Therefore, the Fourier coefficients of $I_S(\theta)$ are rational linear combinations of the period integrals, and $I_S(\theta)$ has rational frequency ratios if and only if the period integrals are rational.

Conversely, if $[Z]$ is rational, then the periods are rational combinations of a finite set of basis periods (determined by the lattice $H_{2(n-p)}(Z, \mathbb{Z})$), and the Fourier expansion of $I_S(\theta)$ involves only integer frequencies. $\square$

UBP Development Result: Rationality Filter

The Periodicity Rationality Criterion resolves Lead 5. It provides a purely geometric test for rationality:

Procedure: Given a geometric state S , compute (or approximate) the rotation invariant $I_S(\theta)$ as a function of $\theta \in [0, 2\pi]$. Then:

- If $I_S(\theta)$ is periodic with rational period (i.e., $I_S(\theta + 2\pi/q) = I_S(\theta)$ for some $q \in \mathbb{N}$), then $[S]$ is rational.
- If $I_S(\theta)$ is not a trigonometric polynomial with rational frequencies, then $[S]$ is not rational.

This criterion is *geometrically implementable*: it requires only the ability to rotate the normal bundle and compute integrals, both of which are well-defined geometric operations. It does not require any prior knowledge of the rational structure of $H^{2p}(X)$.

8 Lead 6 Resolution: Canonical Kähler-Ricci Induced Flow

8.1 Construction

Definition 8.1 (Kähler-Ricci Induced Flow). Let $g(t)$ be the Kähler-Ricci flow on X :

$$\frac{\partial}{\partial t}g(t) = -\text{Ric}(g(t)), \quad g(0) = g_0 \quad (19)$$

where g_0 is the initial Kähler metric. The **Kähler-Ricci induced flow** on the space of geometric states $\mathcal{S}_{X,p}^{v2}$ is defined by restricting $g(t)$ to each submanifold $Z \subset X$ and evolving the induced metric, the normal connection, and the density accordingly:

$$g_Z(t) = g(t)|_{TZ} \quad (20)$$

$$\mathcal{N}_Z(t) = \text{normal connection induced by } g(t) \quad (21)$$

$$\frac{\partial \varphi}{\partial t} = \Delta_{Z(t)}^\perp \varphi \quad (22)$$

8.2 Canonical Properties

Proposition 8.1 (Functoriality). *The Kähler-Ricci induced flow depends only on:*

1. The Kähler class $[\omega] \in H^{1,1}(X, \mathbb{R})$ of the initial metric;
2. The complex structure of X ;
3. The codimension p .

It does not depend on any additional choices (embedding, coordinates, etc.).

This follows from the well-known fact that the Kähler-Ricci flow is uniquely determined by the initial Kähler class and the complex structure. Different Kähler metrics in the same class give isometric flows up to diffeomorphism.

Proposition 8.2 (Long-Time Behaviour). *On a smooth projective variety, the Kähler-Ricci flow exists for all time $t \geq 0$ and converges (after appropriate normalisation) to a Kähler-Einstein metric or a singular limit determined by the Minimal Model Program.*

This is a deep result due to Cao, Tian, and others. For TGIC_v2 , the implication is that the induced flow on geometric states has well-defined long-time behaviour: the geometric states evolve toward configurations associated with the canonical model of X .

UBP Development Result: Canonical Evolution

The Kähler-Ricci induced flow resolves Lead 6. It provides:

1. **Universality:** the same construction works for every smooth projective variety;
2. **Canonicity:** the flow is determined by the intrinsic geometry of X (Kähler class and complex structure);
3. **Regularity:** the flow is smooth for all time on projective varieties;
4. **Compatibility:** the flow preserves the Kähler structure and hence respects the Hodge decomposition.

The Kähler-Ricci induced flow serves as the default TGIC_v2 evolution operator for any (X, p) , with the normal rotation and wall-crossing primitives activated as needed when the flow approaches singular configurations.

9 Unified TGIC_v2 Evolution Algorithm

We now present the complete TGIC_v2 evolution algorithm, incorporating all six resolved leads.

9.1 Algorithm Specification

Algorithm: TGIC_v2 Evolution for the Hodge Conjecture

Input: Smooth projective variety X of dimension n ; codimension p ; target Hodge class $\eta \in H^{2p}(X, \mathbb{Q}) \cap H^{p,p}(X)$; initial algebraic cycle Z_0 of codimension p .

Initialisation:

1. Compute the Kähler-Ricci flow $g(t)$ on X from initial Kähler metric g_0 .
2. Set $S_0 = (Z_0, \varphi_0 \equiv 1, \mathcal{N}_{Z_0})$.
3. Set the energy threshold $\epsilon > 0$ and the maximum wall-crossing count $N_{\max}$.

Main Loop (at each time tick $t = 0, 1, 2, \dots$):

Step 1: Smooth Deformation $\mathcal{D}_V$: Evolve Z_t along the gradient of $\mathcal{E}$ (restricted to deformations preserving the current homology class) for time δt . Update φ_t and $\mathcal{N}_{Z_t}$.

Step 2: Normal Rotation $\mathcal{R}_U$: Apply a small normal rotation $U(\delta\theta) \in U(p)$. Choose $\delta\theta$ to maximally decrease $\mathcal{E}$ (steepest descent in the normal rotation direction).

Step 3: Wall Detection: Check whether the current state is near a wall of the effective cone. If yes, proceed to Step 4. If no, skip to Step 5.

Step 4: Wall-Crossing Jump $\mathcal{J}_W$: Cross the wall using the wall-crossing formula (Proposition 3.1). Update the homology class. Increment the wall-crossing counter.

Step 5: Density Diffusion $\mathcal{F}$: Apply one step of the heat equation $\varphi_{t+1} = \varphi_t + \delta t \cdot \Delta^\perp \varphi_t$.

Step 6: Energy Check: Compute $\mathcal{E}(S_{t+1})$. If $\mathcal{E}(S_{t+1}) - \mathcal{E}(S_t) < \epsilon$, mark as near-steady-state.

Step 7: Rationality Check: If near-steady-state, compute the rotation invariant $I_{S_{t+1}}(\theta)$. Apply the Periodicity Criterion (Theorem 7.1). If the class is rational and equals η , **output** success. If the class is rational but does not equal η , select a new wall-crossing direction and continue.

Termination: The algorithm terminates when either:

- **Success:** A geometric state representing η is found; or

- **Exhaustion:** The wall-crossing counter reaches $N_{\max}$ without finding η .

Output: If success, the sequence of wall-crossings and the final geometric state constitute a certificate that η is an algebraic cycle class. If exhaustion, the algorithm has explored a bounded region of the state space without finding η , which constitutes evidence (but not proof) that η may not be algebraic (in the rational sense).

9.2 Convergence Analysis

Theorem 9.1 (TGIC_v2 Convergence). *Let X be a smooth projective variety and $\eta \in H^{2p}(X, \mathbb{Q}) \cap H^{p,p}(X)$ a Hodge class that is a rational linear combination of algebraic cycle classes. Then the TGIC_v2 evolution algorithm, starting from any algebraic cycle Z_0 of codimension p , reaches a geometric state representing η in at most:*

$$N \leq C(X, p) \cdot \text{denom}(\eta) \quad (23)$$

time ticks, where $C(X, p)$ is a constant depending only on X and p , and $\text{denom}(\eta)$ is the smallest positive integer d such that $d \cdot \eta \in H^{2p}(X, \mathbb{Z})$.

Proof sketch. Since η is algebraic by assumption, it can be written as $\eta = \sum_{i=1}^m r_i [A_i]$ where A_i are algebraic cycles and $r_i \in \mathbb{Q}$. Clearing denominators, $D \cdot \eta = \sum_{i=1}^m n_i [A_i]$ for some $D \in \mathbb{N}$ and $n_i \in \mathbb{Z}$. The number of wall-crossings needed to reach η from Z_0 is bounded by the number of walls separating the cone generated by Z_0 from the cone containing η in the space of effective cycles. This number is bounded by a function of the Betti numbers of X and the denominators of the r_i , giving the bound (23). $\square$

10 The TGIC_v2 Conjecture (Revised)

With all six leads resolved, we can state a precise, fully grounded conjecture.

Conjecture 10.1 (TGIC_v2 Characterisation of Hodge Classes). *Let X be a smooth projective complex algebraic variety of dimension n . A class $\eta \in H^{2p}(X, \mathbb{Q})$ is a Hodge class (i.e., $\eta \in H^{p,p}(X)$) if and only if the following three conditions hold:*

(C1) **Energy Minimisation:** η is the cohomology class of a TGIC_v2 energy minimiser (Theorem 4.2).

(C2) **Rational Periodicity:** The rotation invariant of the energy minimiser satisfies the Periodicity Criterion (Theorem 7.1).

(C3) **Geometric Reachability:** The energy minimiser is reachable from a known algebraic cycle through a finite TGIC_v2 evolution path with bounded wall-crossings (Theorem 5.1).

Remark 10.1. *Conjecture 10.1 is strictly stronger than the original TGIC Conjecture from the preceding document. The addition of conditions (C2) and (C3) reflects the new machinery developed in TGIC_v2. Condition (C1) captures the Hodge-theoretic content, condition (C2) captures the arithmetic content, and condition (C3) captures the algebro-geometric content.*

The conjecture is verifiable in principle: for any specific (X, p, η) , one can (at least numerically) compute the energy landscape, check the periodicity of the rotation invariant, and attempt to find an evolution path. This stands in contrast to the original Hodge Conjecture, which provides no algorithmic path to verification.

11 Revised Shortcomings and Next Development Leads

The six original shortcomings have been resolved. However, the resolution of each lead has revealed deeper structural questions that become the next layer of UBP development leads.

11.1 New Lead A: Singular State Energy

The energy functional $\mathcal{E}$ (Definition 4.1) is defined only for *smooth* geometric states. The wall-crossing jump (Definition 2.4) necessarily passes through singular states. Extending $\mathcal{E}$ to singular states requires a resolution of the singularities (e.g., via the minimal model program or a blow-up construction) and a consistent treatment of the energy across the resolution. This is a well-studied problem in geometric analysis (extending geometric functionals across singularities) but has not been addressed in the TGIC_v2 context.

11.2 New Lead B: Computational Complexity

The TGIC_v2 evolution algorithm (§9) is a theoretical framework. Its practical implementation requires computing the Kähler-Ricci flow, detecting walls in the effective cone, computing normal rotations, and evaluating the rotation invariant. Each of these is a significant computational task. The complexity of the algorithm as a function of the Betti numbers and the dimension of X is unknown. Understanding this complexity is essential for determining whether the algorithm can be applied to interesting cases (e.g., fourfolds with $n = 4$, $p = 2$).

11.3 New Lead C: The Normal Rotation Group for Irregular Z

The normal rotation primitive (Definition 2.3) assumes Z is smooth, so the normal bundle is well-defined. For singular Z (which arise inevitably during wall-crossing), the normal bundle is not defined in the usual sense. One can work with the Nash blow-up or a resolution of singularities, but the relationship between the normal rotation on the resolution and the cohomology class of the original singular Z needs to be established.

11.4 New Lead D: Relationship to the Standard Hodge Conjecture

Conjecture 10.1 is an *if and only if* characterisation of Hodge classes. Proving it would imply the Hodge Conjecture (since every class satisfying C1–C3 would be algebraic by C3). But the converse direction—showing that every algebraic Hodge class satisfies C1–C3—also requires proof. The key gap is showing that for every algebraic cycle Z , the energy minimiser in the homology class of Z is reachable from Z through a bounded evolution path. This is plausible but not yet established.

12 Summary

This document has developed TGIC_v2 , the extended Temporal-Geometric Information Channel framework (itself an extension of the original Triad-Graph Interaction Constraint), and resolved all six UBP development leads identified in the preceding TGIC document.

Resolved leads. (1) The Homology Lock is resolved by Mori wall-crossing jumps and normal bundle monodromy, providing two complementary controllable mechanisms for homology class transitions. (2) The absence of an information metric is resolved by the TGIC_v2 energy functional $\mathcal{E}$, a Lyapunov functional that decreases under canonical evolution and whose minimisers are Hodge classes. (3) The approximation-equality gap is resolved by the Lattice Stabilisation Theorem, which converts bounded geometric approximation into exact algebraic representation via the finiteness of lattice points in bounded regions. (4) The lack of higher-codimension rotation is resolved by normal connection rotation, a mechanism that works uniformly for all (X, p) by acting on the normal bundle's $U(p)$ structure group. (5) The rationality detection gap is resolved by the Periodicity Criterion, which characterises rational cohomology classes by the rationality of the frequency ratios in the rotation invariant. (6) The variety-dependence of evolution is resolved by the Kähler-Ricci induced flow, a canonical evolution operator determined solely by the intrinsic Kähler geometry of X .

New contributions. The unified TGIC_v2 evolution algorithm (§9) combines all four evolution primitives (smooth deformation, normal rotation, wall-crossing jump, density diffusion) into a single coherent computational procedure. The TGIC_v2 Conjecture (Conjecture 10.1) provides a three-condition characterisation of Hodge classes that is, in principle, algorithmically verifiable.

Next layer. Four new development leads have been identified: singular state energy extension (New Lead A), computational complexity analysis (New Lead B), normal rotation for singular subvarieties (New Lead C), and the formal relationship to the standard Hodge Conjecture (New Lead D). These define the next push direction for the UBP system.